

Problem Sheet — IMC Training

June 10, 2026

Problem 1. (IMC 2010, Day 1, Problem 1) Let $0 < a < b$. Prove that

$$\int_a^b (x^2 + 1)e^{-x^2} dx \geq e^{-a^2} - e^{-b^2}.$$

Problem 2. (IMC 2011, Day 2, Problem 3) Determine the value of

$$\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right) \cdot \ln\left(1 + \frac{1}{2n}\right) \cdot \ln\left(1 + \frac{1}{2n+1}\right).$$

Problem 3. (IMC 2014, Day 1, Problem 1) Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\text{trace}(M) = a$ and $\det(M) = b$.

Problem 4. (IMC 2010, Day 1, Problem 3) Define the sequence $x_1, x_2, \dots$ inductively by $x_1 = \sqrt{5}$ and $x_{n+1} = x_n^2 - 2$ for each $n \geq 1$. Compute

$$\lim_{n \rightarrow \infty} \frac{x_1 \cdot x_2 \cdot x_3 \cdots x_n}{x_{n+1}}.$$

Problem 5. (IMC 2011, Day 1, Problem 2) Does there exist a real 3×3 matrix A such that $\text{tr}(A) = 0$ and $A^2 + A^t = I$? (Here A^t is the transpose of A and I is the identity matrix.)

Problem 6. (IMC 2010, Day 2, Problem 3) Denote by S_n the group of permutations of $(1, 2, \dots, n)$. Suppose that G is a subgroup of S_n such that for every $\pi \in G \setminus \{e\}$ there exists a unique $k \in \{1, 2, \dots, n\}$ with $\pi(k) = k$. Show that this k is the same for all $\pi \in G \setminus \{e\}$.

Problem 7. (IMC 2014, Day 2, Problem 3) Let $f(x) = \frac{\sin x}{x}$ for $x > 0$, and let n be a positive integer. Prove that

$$\left| f^{(n)}(x) \right| < \frac{1}{n+1},$$

where $f^{(n)}$ denotes the n -th derivative of f .

Problem 8. (IMC 2011, Day 1, Problem 4) Let $A_1, A_2, \dots, A_n$ be finite, nonempty sets. Define the function

$$f(t) = \sum_{k=1}^n \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} (-1)^{k-1} t^{|A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}|}.$$

Prove that f is nondecreasing on $[0, 1]$.

Problem 9. (IMC 2010, Day 1, Problem 5) Suppose that a, b, c are real numbers in the interval $[-1, 1]$ such that

$$1 + 2abc \geq a^2 + b^2 + c^2.$$

Prove that for all positive integers n ,

$$1 + 2(abc)^n \geq a^{2n} + b^{2n} + c^{2n}.$$

Problem 10. (IMC 2016, Day 2, Problem 10) Let A be an $n \times n$ complex matrix whose eigenvalues all have absolute value at most 1. Prove that

$$\|A^n\| \leq \frac{n}{\ln 2} \|A\|^{n-1}.$$

Here $\|B\| = \sup_{\|x\| \leq 1} \|Bx\|$ and $\|x\| = \sqrt{\sum_{i=1}^n |x_i|^2}$.